

Modeling method and application of BIM in design stage of long-span suspension bridge

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Abstract:With the continuous promotion and development of building information model (BIM) in the construction field, BIM technology is widely used in various fields. As the carrier of building information, BIM model is the basis of application of BIM technology. Due to the special structure type and complex shape of long-span suspension bridges, it is difficult to establish BIM model and to carry out subsequent application. This paper will use BIM technology to carry out parametric modeling on part of the main structure and complex components of the bridge in the design stage of Taihong Yangtze River Bridge, and its application of digital delivery, forward plotting and engineering measurement will be studied. It provides practical experience for BIM Technology in the design stage of long-span suspension bridge.

Keywords:*BIM; Parametric; Suspension bridge; Modeling*

1. Introduction:

Nowadays, building information technology (BIM) has been recognized by the bridge engineering industry. Scholars and engineers have studied the application of BIM Technology in the bridge design stage. In China, Fuchun Song et al.^[1] established a complete cable-stayed bridge model of Yongchuan

Yangtze River Bridge by using Revit, which improved the modeling speed of the bridge; Yi Wang et al. [2] took Shanghai Beihong road overpass project as the basis and used CATIA as the secondary development method to prepare automatic modeling plug-in for steel box girders, realizing the rapid modeling of steel box girders. In other countries, the Grandpa Bridge in Helsinki, Finland, USA was constructed using BIM technology throughout, accessing the composite models by sharing Tekla models, enabling each participant to use the constantly updated models; South Korea K.M. lee et al. [3] applied BIM 3D steel bar modeling on the design and construction of box girder bridge, which reduced the cost.

There are few researches on the fine modeling of long-span suspension bridge in the industry, mainly due to the special type of bridge structure, complex modeling and so on. Therefore, this paper attempts to apply BIM Technology to the modeling of long-span suspension bridge in the design stage, carrying out the following research: using software to model the complex components of the main structure of the bridge, and studying the application of BIM Technology in digital delivery, forward plotting and engineering measurement.

2. Selection of application software in the design stage of Taihong Yangtze River Bridge

At present, the BIM design software commonly used comes from four companies, Bentley, Tekla, Autodesk and Dassault [4]. As shown in figure 1, the BIM software classification of each platform is shown. The first, the Bentley series of design software from the United States, it has a good market in factories and infrastructure. In recent years, it has demonstrated its powerful 3d modeling capabilities in roads and Bridges; The second, Tekla structures software of Tekla company. Tekla structures is a steel structure detail design software, which can automatically to generate steel structure details and various reports through the 3D model created; The third, Revit software of AutoCAD company, which has great advantages in large-scale building modeling, and also involves structural and electromechanical industry modeling; The fourth, Catia software of French Dassault company, it has been in a monopoly situation in the mechanical aspect. The reason is that this software has powerful curved surface design function and parametric design function, but it is seldom applied for the civil engineering, so it has a very broad prospect.

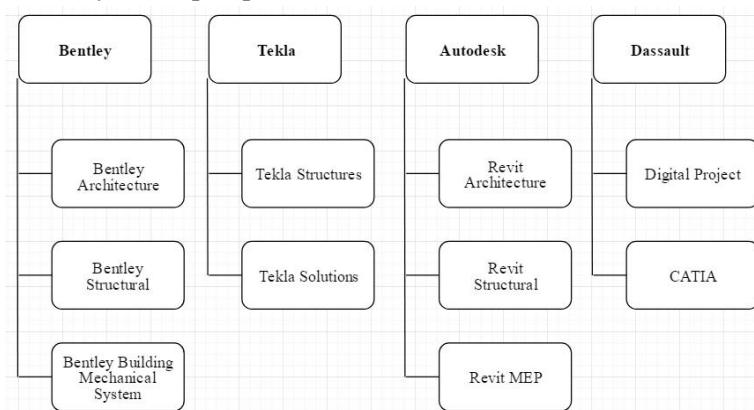


Figure 1. BIM software classification of each platform.

As for the choice of BIM technology design software, different structural types can be modeled with different software. According to the application feedback on various software operating platforms

at ordinary times, the mutual cooperation between the software can often achieve twice the result of half the effort ^[5-6]. The first three software platforms are selected in this project. CivilStation Bridge is a software mainly used in bridge structure design by Bentley platform, which provides functions such as line-based bridge span arrangement, parametric modeling of bridge concrete structure, customized parametric component template, etc; Tekla Structures is a commonly used steel structure design software, which is highly specialized and widely used. Tekla Structures is widely used as a tool for Steel Structure Deepening in steel structure processing plants; The auxiliary structure of the main tower crown is constructed on site with relatively few participants and less work, so the ProStructures software can be used for modeling; Revit is the basic software used in structural design and parametric design in Autodesk series software. Its greatest advantage is that it realizes the transformation from two-dimensional design to three-dimensional design, at the same time, it further realizes parametric, precise and intelligent design, and realizes the functions that traditional two-dimensional design software such as automatic statistics of engineering quantity and visual construction simulation do not have. Table 1 shows the main application software of BIM model of Taihong Yangtze River Bridge.

Table 1 Main application software of BIM models on Taihong Yangtze River Bridge .

Modeling software	Model object	Technical characteristics
CivilStationBridge	Bridge Structure	It provides functions such as bridge span layout based on lines, customized parametric component template, etc. The user-defined parametric component functions greatly to enrich the component templates that meets the design characteristics of the design institute.
TeklaStructures	Reinforced Beam	Teklastructures is a commonly used software for steel structure design. It is highly specialized and widely used. It is widely used in steel structure processing plants, which is also beneficial for the next stage of intelligent manufacturing.
ProStructures	Tower Crown	In order to compare the application prospect of practical engineering, the main tower crown is modeled by ProStructures, which is parallel to Teklastructures. In addition, the steel structure is also modeled by ProStructures.
Revit	Cable Clamp	By changing the size design parameters of cable clamps in Revit software family types, the automatic adjustment of cable clamp structure and the automatic generation of parametric design models of different types of cable clamps can be realized.

3. Establishment of bridge BIM model

3.1. BIM modeling of bridge main structure

(1) BIM model of T-beam and T-structure of approach bridge

CSB is used to model the T-beam of the approach bridge. The template for its bridge component is generated by the self-contained PCL script language, and its process is as follows: The PCL script is interpreted to generate a template. The template generates a complex shape according to the specified parameters, and the program automatically assigns attributes to the shape. The built-in component template basically meets the requirements of current use, and the PCL file needs to be modified or newly created for special needs. The bridge deck system is created by the CSB automatically calling the PowerCivil corridor function. You can also select the appropriate corridor sections template or custom section template to create the corridor model including the bridge deck system.

Top elevation of T-frame upper structure changes of route and bottom elevation changes with function curve. Internal structure dimensions are complex. For this kind of structure of complex geometry size, it is advisable to use the function of "placing the main beam of corridor" of CSB to design the parameterized corridor section and assign the parameters of each section. Firstly, create a customized Gallery cross-section template, as shown in Figure 2. The main task of creating the template is to organize the logical relationship between the control points of the cross-section so that a defined cross-section size can be controlled with as few parameters as possible. Secondly, the key points of the template should be named, including the constraints on point characteristics, which should reflect the constraint relationship between each key point and clearly express the meaning of constraints; Finally, the T structured is created with the default parameter value. Assign values of each parameter in the excel table, and import the corridor object parameter constraint table according to the rules required by the software, so as to create an accurate T-structure superstructure, as shown in Figure 3.

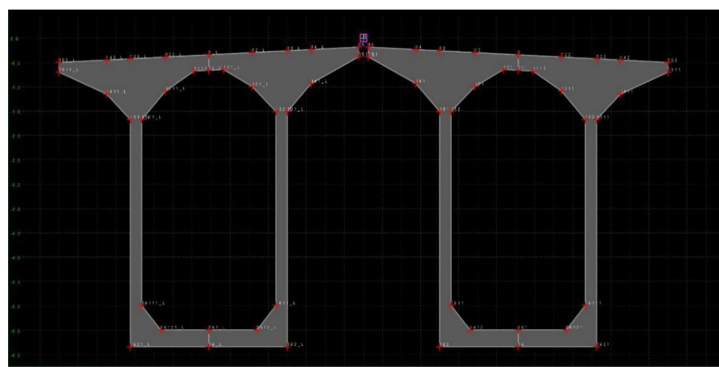


Figure 2. Logic designed of key point parameters of T-section.

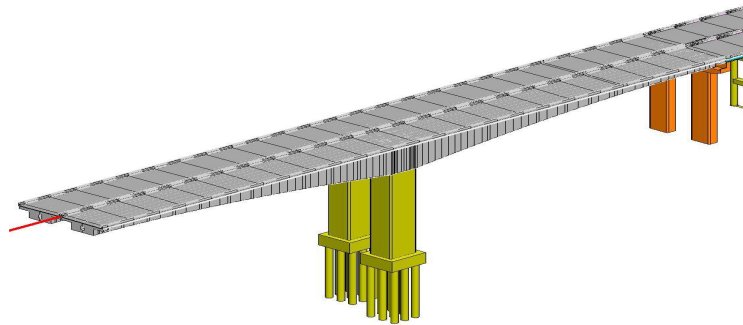


Figure 3. T-frame model of approach bridge.

(2) BIM model of main bridge tower

CSB provides powerful component parameterization, Parameterization provides the creation of new component types with PCL parameterized scripts, For commonly used or software without built-in component form, component templates can be created by this method, so as to facilitate reuse of later projects. PCL script is prepared for the main bridge tower of the Taihong Bridge project, which achieves better application effect. The model size of the two bridge towers can be quickly modified by modifying the model parameters, as shown in Figure 4.

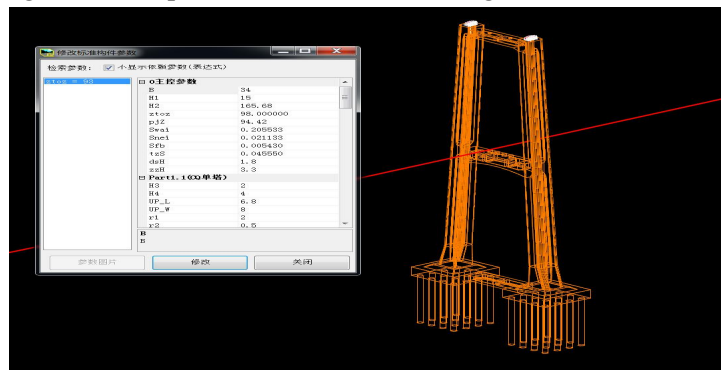


Figure 4. Adjustment of structural parameters of bridge tower.

(3) BIM model of steel structure

ProStructures and Teklastructures are two feasible options for steel structure modeling at this stage. The structural members of Taihong project are mainly steel box girders and bridge tower crown, The auxiliary structure of the main tower crown is field construction, which involves relatively few participants and works amount. Therefore, ProStructures will be used for the steel structure of the tower crown; In order to better to connect various parts of work, Tekla Structures is widely used as a tool for steel structure deepening in steel processing plants^[7], The steel box girders section of the main bridge is to be created using Tekla Structures. Figure 5 and Figure 6 respectively show the auxiliary structure of the tower crown and the steel box girder model.

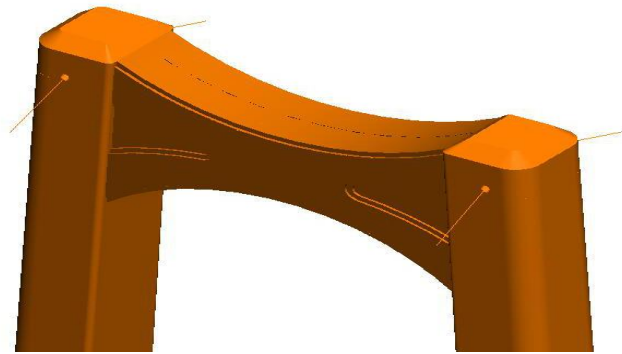


Figure 5. Auxiliary structure of bridge tower crown.

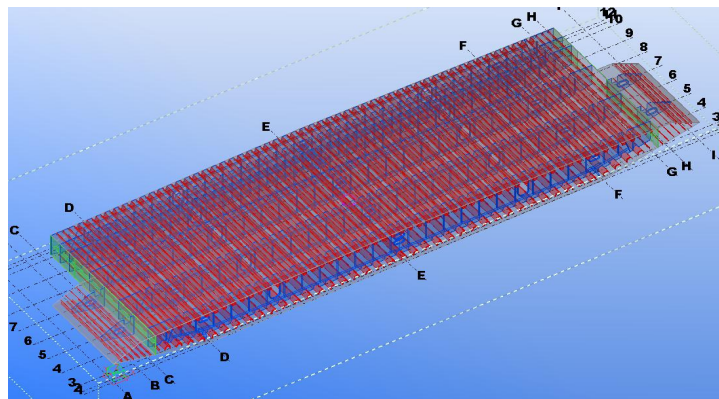


Figure 6. Beam segment model.

3.2. *BIM modeling on large and complex components of bridges*

(1) BIM model of reinforcement construction

The steel bar is modeled by Proconcrete module in ProStructures. Due to the large number of steel bars, in order to make the modeling and the use of the model more convenient and efficient, each type of steel bar needs to be set with a separate soil layer. The structural shape of Taihong bridge is more complex than that of conventional bridge, and the shape of reinforcement is also more complex correspondingly. Figures 7 and 8 are respectively the structural models of steel bar for bridge tower abutment and the structural models of steel bar for bottom cross member. Curved reinforcing bars, such as 3D helical reinforcing bars in pile foundations, are particularly expensive for computer resources. ProStructures software does not provide an excellent solution to such complex reinforcements. It will take a lot of work to create the reinforcement structure. For the optimization of complex structure reinforcement and curve reinforcement resource occupation, it may need the development of professional software to obtain a better solution.

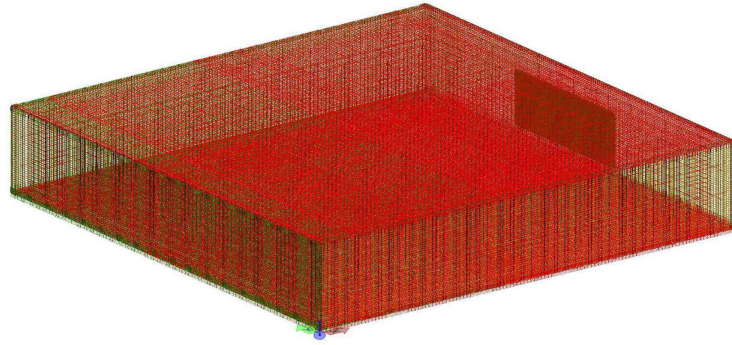


Figure 7. Reinforcement structure of bridge tower bearing platform.

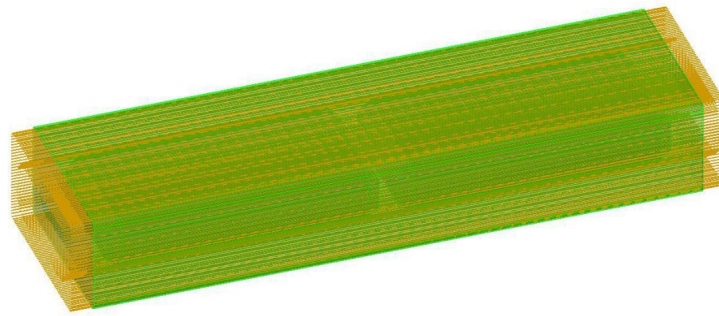
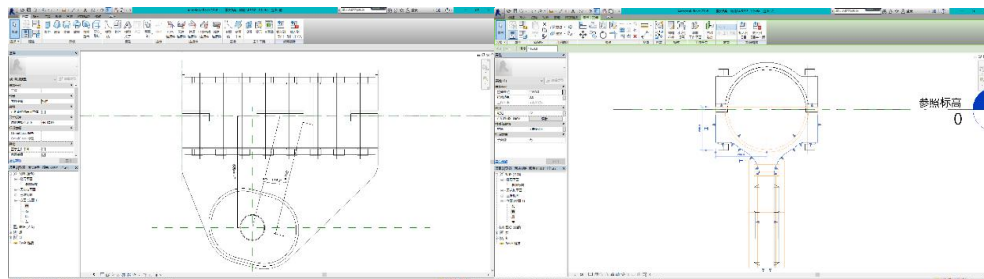


Figure 8. Reinforcement structure of bottom crossbeam.

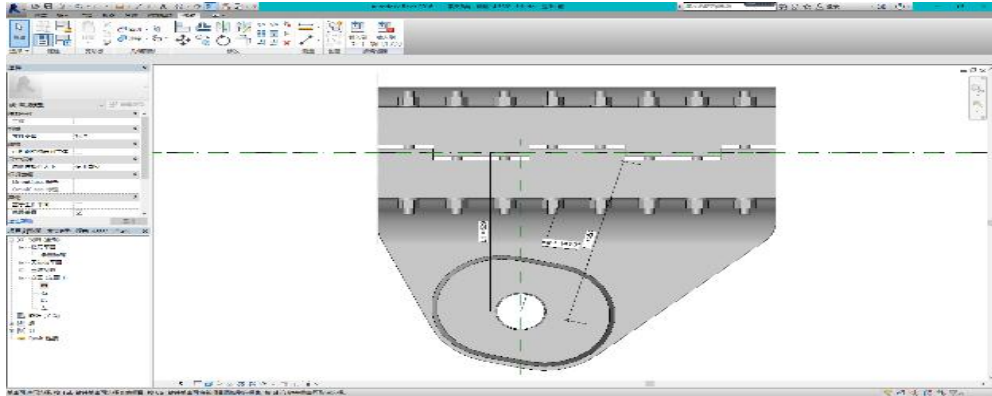
(2) BIM model of cable clamp

There are a large number of complex components in suspension bridges, such as cable clamps. BIM modeling can show the spatial characteristics of components more intuitively and avoid the 3D size mismatch caused by the plan. The parametric design of the general structural model of bridge is a very powerful function of Revit software, which uses the 3D reference line or 3D reference planed of Revit software to locate the parametric profile. Through the functions of dimension, parameter constraint, association reference and dimension constraint, the parametric design of component profile is carried out. Taking the cable clamp of suspension bridge cable system as an example, the parametric design of cable clamps contour and the setting of control parameters are shown in Figure 9.



(a)

(b)



(c)

Figure 9. Parameterized design of cable clamp profile and setting of control parameters.

Revit is one of the most representative design software under BIM platform. Parametric design, component relevance design, parametric driven shape design and collaborative design are the main features of the software. The parametric design operation interface of the software is very clear^[8]. Based on the parametric design of the cable clamp profile, the solid model or hollow model is created by extruding, rotating, fusing, lofting, mirroring, moving, array and other solid modeling commands. It can help to complete the design of the cable clamp entity model with the functions of cutting, connecting, adding parameter expression, Boolean operation, logical operation, etc., and further verify the success of the parameter setting of the parametric design model through the modification of structural design parameters and the automatic adjustment of the parametric design model. The 3D effect of parametric cable clamp is shown in Figure 10.

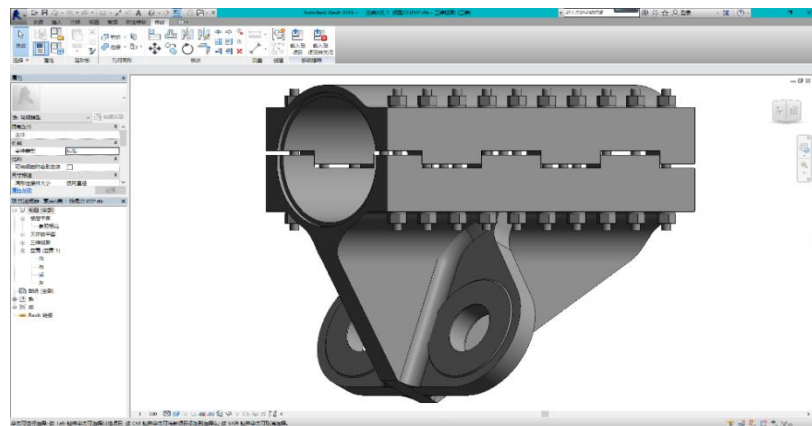


Figure 10. Three dimensional effect display of parametric cable clamp.

3.3. Three dimensional real scene model of BIM

The BIM model established above is overlapped with the high-precision tilt photography model. The tilt photography model is a kind of terrain model data generated by high-precision aerial survey photos, with the highest accuracy of 0.05m. Due to the ultra-high precision of the real scene model, the real situation of the surrounding areas of the project can be fully displayed, which greatly facilitates the

early construction work such as land acquisition and demolition, land measurement, and sensitive area avoidance, and defines the division of influencing factors. Figure 11 shows the real scene model of Taihong Yangtze River Bridge.

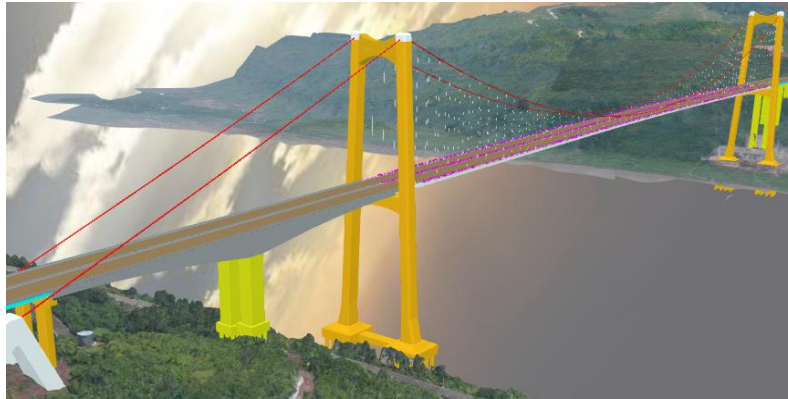


Figure 11. Real scene model of Taihong Yangtze River Bridge.

Through the 3D real scene model, we can see the topography and actual situation of the whole area from multiple angles, which has the effect of on-site inspection. In this way, the 3D real-world model can help the designer to understand the site conditions, avoid unfavorable sections, reduce the impact on existing routes, and make intuitive judgment and scheme comparison and Analysis on bridge location selection. The application of 3D scene model based on BIM in the preliminary design stage improves the overall route design, comparison and selection efficiency and design accuracy of designers.

4. Application research of BIM model

4.1. Application research of digital delivery

Data management of engineering project firstly needs to be the requirement of basic project management, to put forward specific requirements for digital engineering and physical actual engineering, and to define and rule the data generated and output at each stage. In the process of management, a series of methods and tools are needed to ensure that project execution meets expectations. As the carrier of bridge structure visualization and building information, BIM should be closely connected with actual buildings to maximize real-time management quality and realize real-time digital delivery. Real-time model viewing and detail browsing through mobile side is an important way of data delivery, while all BIM platforms running on PC host are difficult to meet the real-time visual management on site. Using Navigator, the model can be accessed and viewed on mobile side such as smartphone or tablet in real-time. Figures 12 and 13 are BIM models in PC and BIM models in IPAD respectively. The main technical methods are as follows:

- (1) Open BIM model object with MicroStation CE version;
- (2) Publish the model as i-model file in MicroStation;
- (3) Use iPad to install Navigator mobile terminal;
- (4) Loading the i-model file of BIM object into the Navigator of mobile terminal can realize the visual operation of the model.

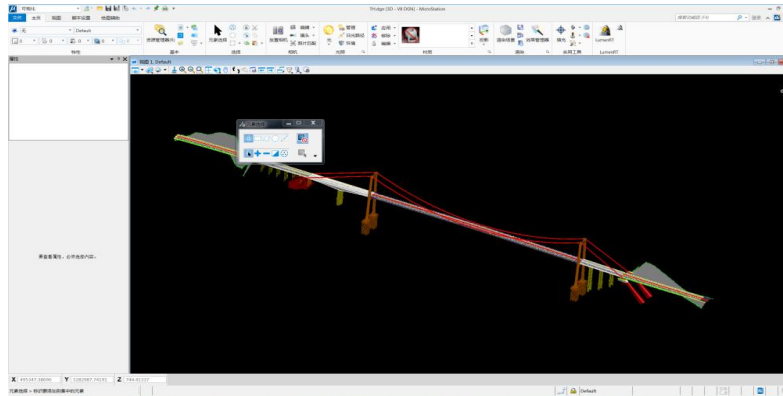


Figure 12. The BIM model objects in the PC end.



Figure 13. shows the BIM model in IPAD.

4.2. Drawing of BIM model

Main view, top view and left view are three basic views that can correctly reflect the length, width and size of the object. They are an abstract expression customarily used for the geometrical shape of the object of engineering circles and the basis of current two-dimensional drawings. At present, two-dimensional drawings are the main delivery documents in Chinese construction industry. Therefore, all processes of bridge construction in China now revolve around "drawings". General bridges rely on two-dimensional drawings, while special bridges rely mainly on two-dimensional drawings and supplemented by three-dimensional drawings^[10].

Under the application frame of BIM, the design concept gradually changes to "three-dimensional mainly, two-dimensional supplementary". Some BIM software can automatically output two-dimensional drawings. The three-dimensional model can not only satisfy the requirements of complex and changeable spatial expression, but also generate two-dimensional drawings, which can not only enable the owner to backup, but also satisfy the requirements of a large number of construction units for two-dimensional drawings for a long time.

The model of reinforced concrete box girders of Taihong Bridge was created by Tekla Structures software. Through reasonable planning of component, part attributes and naming during the modeling process, and setting of drawing template and variation, the fine drawing of the model was completed. Figures 14 and 15 show the pattern template settings and part drawings respectively. This software

supports filtering options, enabling drawings for a large project to be produced in multiple batches, enabling the design unit to design phase II while the factory can produce phase I components first, reducing the project cycle^[7].

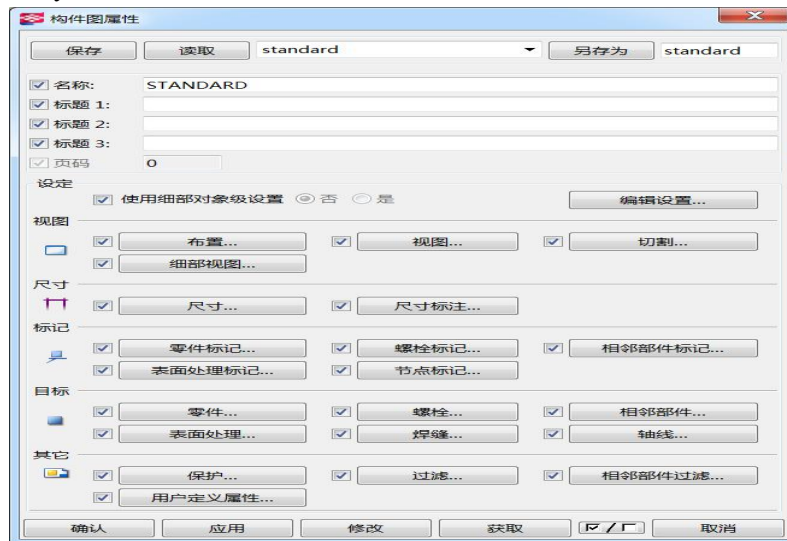


Figure 14. Setting of drawing style template.

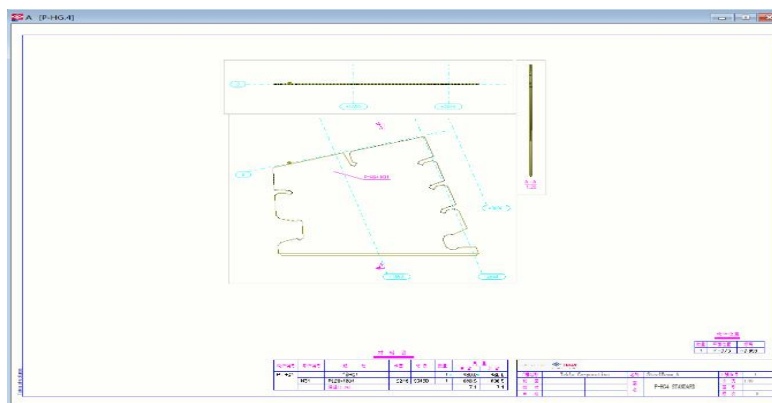


Figure 15. Part drawing.

4.3. Engineering measurement

Engineering measurement is a unified summary and statistics of the quantity of materials involved in the design, which is convenient for the overall management before and during the construction. Although engineering measurement is carried out after the completion of modeling according to the workflow, in fact, the planning of measurement work should be fully considered in the modeling process. BIM technology can meet the engineering measurement and pricing requirements for large volume and special special-shaped projects ^[11]. On the basis of appropriate modification of the engineering cost management rules, the BIM model based on accurate information will quickly provide the construction project quantity information.

Tekla has created a steel box girder model with abundant attributes, and has provided abundant engineering metering to report templates by itself. If the requirements are not satisfied, the template can also be customized. Reports are informative and highly customizable. Figures 16 and 17 are the

diaphragm model of reinforced concrete box girder and the metering list of diaphragm members respectively. The software automatically numbers the components according to the model settings, and automatically identifies the same components in the model according to the set error range from normalization statistics.

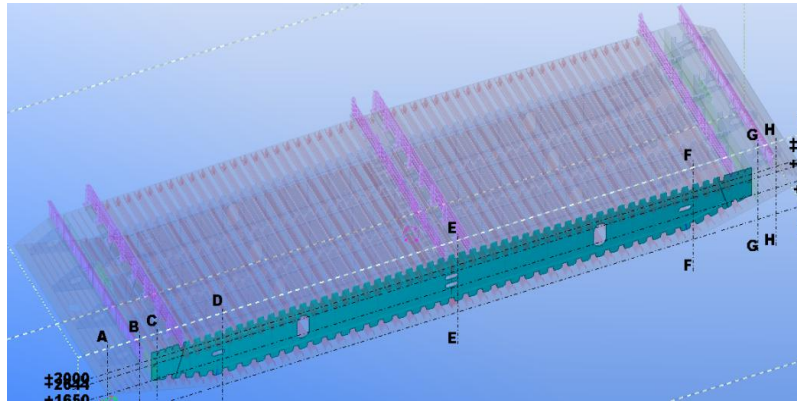


Figure 16. diaphragm model.

构件编号	数量	型材	表面积 (m ²)	重量 (kg)
NS2	2	BL140*10	0.3	9.6
M201	2	PL12*400	0.3	15.2
P-HG1	2	PL12*2440	63.5	2951.7
P-HG2	2	PL12*550	13.4	604.9
P-HG3	2	BL172*10	6.5	204.8
P-HG4	1	PL20*1894	6.2	472.9
P-HG5	1	PL20*1894	6.2	472.9
P-HG6	2	BL140*10	0.4	170.9
P-HG7	2	BL120*10	0.5	18.1
P-HG8	2	BL120*10	0.5	17.8
P-HG9	2	BL120*10	0.5	17.6
P-HG10	2	BL120*10	0.5	17.4
P-HG11	2	BL120*10	0.5	17.2
P-HG12	2	BL120*10	0.5	16.9
P-HG13	2	BL120*10	0.5	16.7
P-HG14	2	BL120*10	0.5	16.5
P-HG15	2	BL120*10	0.5	16.3
P-HG16	2	BL120*10	0.4	16.0
P-HG17	2	BL120*10	0.4	15.8
P-HG18	2	BL120*10	0.4	14.2
P-HG19	2	BL120*10	0.3	11.7
P-HG20	2	BL120*10	0.3	11.1
P-HG21	2	BL120*10	0.9	34.1
P-HG22	2	BL120*10	0.5	16.6
P-HG23	2	BL120*10	0.5	17.0
合计	48	构件数	204.0	9462.1

Figure 17. measurement list of diaphragm member.

5. Conclusion

Based on the Taihong Yangtze River Bridge project, this paper applies BIM technology to the design stage of long-span suspension bridge. The corresponding software of each platform is used to parameterize the main structure and complex components of the bridge. The problem of model sharing between different software, mobile terminal and PC terminal and different data formats of BIM model is solved, and the digital delivery is preliminarily realized. The forward drawing based of BIM model are studied and the BIM model is mastered. The method of directly generating 2D anisotropic map by 3D model is studied; engineering measurement based on BIM model is studied; and the method of directly generating engineering measurement list of BIM information management database is mastered.

BIM technology implementation is the second change in construction engineering design, for the project participants work together provide a platform for the exchange, reduce design cost and rework

risk, effectively improve the quality of project engineering designs, reduce the construction cost, but the BIM technology in the process of bridge design without strong pertinence, and there are a variety of problems, which requires people to gradually perfect the modeling software, accumulated certain experience to solve.

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